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MILITARY SPECIFICATION

TUMBLER-DRYER LAUNDRY, STEAM, AND ELECTRIC MODIFIED COMMERCIAL (SHIPBOARD)

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all departments and agencies of the Departments of Defense.

1. SCOPE

1.1 Scope. This specification covers 50- and 100-pound dry weight capacity modified commercial laundry tumbler-dryer machines for Navy shipboard use.

1.2 Classification. The tumbler-dryers shall be of the following types, classes, and styles, as specified (see 6.2).

Type I- Steam heated
- Class A - 50-pound dry weight
  - Style 1 - Unsectionalizes
  - style 2 - Sectionalized
- Class B – 100-pound dry weight
  - Style 1 - Unsectionalized

Type 11 - Electrically heated
- Class A - 50-pound dry weight
  - Style 1 – Unsectionalized
  - Style 2 - Sectionalized

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3510

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications, and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed m the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

PPP-F-320	Fiberboard: Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes

MILITARY

MIL-E-917	Electic Power Equipment, Basic Requirements (Naval Shipboard Use)
MIL-L-3153	Laundry and Dry Cleaning Machinery and Equipment, Packaging of
MIL-M-17060	Motors, 60-Hertz Alternating Current, Integral-Horsepower, Ship-board Use
MIL-L-19140	Lumber and Plywood, Fire-Retardant Treated

STANDARDS

MILITARY

MIL-STD-108	Definitions of and Basic Requirements for Enclosures for Electic and Electronic Equipment
MIL-STD-167-1	Mechanical Vibrations of Shipboard Equipment (Type I – Environmental and Type H - Internally Excited)
MIL-STD-461	Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of

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DOD-STD-1399, Section 300 — Interface Standard for Shipboard Systems Electric Power, Alternating Current (Metric)

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 191 11-5094.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, section II - Material Specifications, Section VIII - Rules for Construction of Pressure Vessels

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 525 — Standard Specification for General Requirements for Steel Sheet, Zinc-Coated (Galvanized) by the Hot Dip Process (DOD adopted)

B 61 — Standard Specification for Steam or Valve Bronze Castings (DOD adopted)

B 62 — Standard Specification for Composition Bronze or Ounce Metal Castings (DOD adopted)

B 117 — Standard Method of Salt Spray (Fog) Testing (DOD adopted)

D 3359 — Standard Methods for Measuring Adhesion by Tape Test (DOD adopted)

D 3363 — Standard Test Method for Film Hardness by Pencil Test

F1166 — Human Engineering Design for Marine Systems, Equipment and Facilities

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS, INC. (IEEE)

Standard 45 Recommended Practice for Electric Installations on Shipboard

(Application for spies should be addressed to the Institute of Electrical and Electronics Engineers, 345 East 47th Street New York, NY 10017.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

National Electrical Code 1987 Edition

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

23 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.5) in accordance with 4.4.

3.2 Materials. Materials shall be as specified in 3.2.1 through 3.2.7.

3.2.1 Structure. Frame, base, and strength members shall be constructed of malleable iron, ductile iron, or steel. Gray cast iron or other brittle materials shall not be used for these parts. Gear reducers and bearing housings are not strength members unless manufactured as an integral part of the frame.

3.2.2 Housing. Housing shall be constructed to encase insulation and finished for corrosion resistance with internal stiffening members (see 4.4.3.8). Interior surfaces shall be finished smooth to the touch and shall not allow lint buildup.

3.23 Heating elements. Heating elements shall be as specified in 3.2.3.1 and 3.2.3.2.

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3.2.3.1 Steam heating coils, type I. Heating coils on type I tumbler-dryers shall be made and constructed in accordance with the ASME Boiler and Pressure Vessel Code, sections II and VIII.

3.2.3.2 Electric tubular heating elements, type II. Tubular heating element sheath material on type II tumbler-dryers shall be made of monel-400 with hermetic end seals or material having equivalent corrosion resistance in a marine atmosphere and maximum operating temperature. The spiral wound resistor wire shall be made of 80 percent nickel and 20 percent chromium and suspended in the dielectric that shall be composed of compressed magnesium oxide.

3.2.4 Thermal insulation. Insulation shall be flame retardant, nontoxic, unsettling, and shall not flake, crumble, or enter the drying airstream during maintenance and operation of the tumbler-dryer. Equipment shall be completely free from asbestos or asbestos treated materials.

3.2.5 Cylinder basket. The cylinder basket shall be made of hot-dipped, galvanized sheet steel in accordance with ASTM A 525, or other corrosion-resistant materials with equal or better corrosion resistance, material strength, and heat strength characteristics. Surface coatings or finishes shall not stain the clothing within the cylinder basket.

3.2.6 Pipes, fittings, valves, and connections. Pipes, fittings, valves, and connectins shall be as specified in 3.2.6.1 through 3.2.6.3.

3.2.6.1 Pipes. Pipes shall be made of corrosion-resistant materials as specified in 3.2.3.1.

3.2.6.2 Fittings Fittings shall be made of bronze in accordance with ASTM B 61 or ASTM B 62.

3.2.63 Valves. Valves shall be made of cast bronze m accordance with ASTM B 61.

3.2.7 Restricted materials. The tumbler-dryer shall be free from asbestos or cadmium parts or coatings in any material form.

3.2.8 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term “recovered materials” means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified

33 Configuration. The configuration of the tumbler-dryer shall be as specified in 3.3.1 through 3.3.8.

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33.1 General. The tumbler-d~er shall be a reversing, open end, once through, end loading machine constructed so the heated air shall not circulate back through the cylinder. Components shall be arranged so that side clearance shall not be required for installation or maintenance, permitting tumbler-dryers to be juxtaposed.

3.3.2 Capacity. The tumbler-dryer shall adequately hold a load of cotton cloth in accordance with 4.4.3.2 not less than 50 pounds dry weight for class A tumbler-dryers and 100 pounds dry weight for class B tumbler-dryers.

3.3.3 Deck mounting. Tumbler-dryer deck mounting shall be in accordance with figures 1 and 2. No other attachment points shall be permitted except for air ducting, electrical cabling, and steam lines. Mounting pads shall be secured to an area of the tumbler-dryer cabinet structure to ensure adherence under ship motion and vibration. Structures penetrated by mounting bolts shall be made of high grade steel. When subjected to shipboard accelerations of 31g's vertically and 25g's horizontally through the center of gravity of the machine, the feet, legs, and mounting frames shall not exceed the yield point of their materials.

3.3.4 Volumetric dimensions. The tumbler-dryers shall not exceed overall, fully assembled dimensional limits specified in table I.

TABLE I. *Maximum volumetric dimensions of equipment.*

Class	Height	Width	Depth
A	80	40	52
B	90	50	68

33.5 Weight limits. The maximum allowed weight limit on class A tumbler-dryers is 900 pounds-force and on class B tumbler-dryers 1,650 pounds-force

33.6 Sectionalization. Style 2 tumbler-dryers shall be constructed to be disassembled, transported, and reassembled within the ship without hot work, such as welding and brazing. Major subassemblies that bolt together shall be preferred. Limiting dimensions for each major subassembly shall be 50 inches high, by 25 inches wide, by 36 inches deep. Subassemblies shall be match-marked to assist in field reassembly once in the designated space.

3.3.7 Controls, indicator lights and audible signals. Controls, indicator lights and audible signals shall be as specified in 3.3.7.1 through 3.3.7.5.

33.7.1 General criteria. Controls shall be selected and integrated into the dryer m order to meet the applicable human engineering requirements specified in ASTM F1 166.

3.3.7.1.1 Distribution of work load. Controls shall be selected and distributed so that none of the operator's limbs will be overburdened.

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3.3.7.1.2 Stops. Stops shall be provided at the beginning and end of the range of control positions if the control is not required to be operated beyond the indicated end positions or specified limits

3.3.7.1.3 Consistency of movement- Direction of control movement shall be consist with the related movement of an associated display or equipment component. Movement of a control forward, clockwise, to the right, or up, or pressing a control shall turn the equipment or component on or cause the quantity to increase.

3.3.7.1.4 Operator-control orientation. Controls shall be oriented with respect to the operator.

3.3.7.1.5 Grouping. Controls that function in sequential operation necessary to a particular task or that operate together shall be grouped together along with their associated displays. When several steps of a sequence are selected by one control, the steps shall be arranged by order of occurrence to minimize control movements and prevent cycling through unnecessary steps. Cycling through the control's ON or OFF position shall be avoided.

3.3.7.1.6 Sequential operation. Where sequential operations follow a fixed pattern, controls shall be arranged to facilitate operation (for example, in a pattern left-to-right and top-to-bottom, like a printed page).

3.3.7.1.7 Location of primary controls. The most important and frequently used controls shall have the most favorable position with respect to ease of reaching and grasping (particularly rotary controls and those requiring tine settings). When a transilluminated indicator is associated with a control, the indicator light shall be so located that it may be associated with the control without error and shall be visible to the operator during control operation.

32.7.1.8 Spacing. Minimum spacing between controls shall be as specified in ASTM F1 166. Spacing between a control and any adjacent obstruction shall be as shown on the figures in ASTM F1166. The spacing between adjacent edges of simple, round indicator light fixtures shall permit unambiguous labeling, signal interpretation, and convenient bulb removal.

3.3.7.1.9 Relation to display. When color-coding must be used to relate a control to its corresponding display, the same color shall be used for both the control and the display.

3.3.7.1.10 Control panel contrast. The color of the control shall be black or gray and shall provide contrast between the panel background and the control.

33.7.1.11 Labeling of controls Controls and displays shall be appropriately and clearly labeled with label plates that contain the basic information needed for proper identification, utilization, actuation, or manipulation of the element. By using simple words and phrases, the labes shall convey verbal meaning in the most direct manner. Each control and display shall be

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labeled according to function, and the following criteria shall apply:

a. Similar names for different controls and displays shall be avoided.

b. Instruments shall be labeled m terms of what is being measured or controlled, taking into account the user and purpose.

c. Control labeling shall indicate the functional result of control movement (for example, ON or OFF). Such information shall be visible during normal operation of the control

d. Labels shall be placed above the controls and displays they describe and shall be oriented horizontally so that they may be read from left to right. When the panel is above eye level and if label visibility will be enhanced, labels may be located below the controls.

e. The units of measurement (for example, degrees Fahrenheit (“F) or minutes) shall be labeled on the panel.

f. Lettering shall be black on a light background.

3.3.7.1.12 Prevention of accidental actuation. Controls shall be constructed and located so that they are not susceptible to being moved accidentally (for example, controls shall be provided with resistance such as viscous or coulomb friction, spring-loading, and so forth).

3.3.7.1.13 Coding. Simple indicator lights shall be coded to conform to the requirements specified m ASTM F1 166.

33.7.1.14 Intensity of audio signals. Audio warning signals shall not be so intense that discomfort or ringing in the ears is caused.

3.3.7.2 Selective temperatures. Means shall be provided for manually setting the temperature best adapted to the material being dried. A temperature indicator shall be provided for observation of the actual temperature. A thermostatic control, which operates the steam solenoid valve on type I tumbler-dryers or the electric heater control on type II tumbler-dryers, shall be set and maintain two selected temperatures of 185 and 155 ‘F.

33.73 Timed drying cycle. A timer shall be provided with a range of O to 60 minutes by which the operator can select the drying time. An indicator light shall be provided to indicate normal drying operation.

33.7.4 Cooldown phase. A solenoid valve and relay shall be provided for cutoff of steam to heating coils on type I or electrical power to the heating elements of type II tumbler-dryers during the cooldown phase respectively. Cooldown shall be automatically activated until cylinder tem-

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perature drops to 135 "F. An indicator light shall be provided which is activated for the duration of the cooldown phase. The timer shall also activate an audible signal at the end of the cooldown cycle The audible signal shall continue until manually turned off.

3.3.7.5 Temperature limit control. Steam-to-heating coils on type I tumbler-dryers and electrical power-to-heating elements on type II tumbler-dryers shall be under control of primary and backup heat sensors and an airflow sensor. The airflow sensor shall be constructed to prevent intermittent actuation when exposed to shipboard vibration. Sensors shall interrupt steam flow by means of a solenoid valve or Power-t-heating elements through a relay. Reset shall be accomplished automatically or by the start button as determined by human engineering considerations (see ASTM F1 166).

3.3.8 Human factors engineering. Human engineering criteria, principles, and practices, as specified in ASTM F1 166, shall be used in the configuration of the laundry tumbler-dryer where specified throughout this document. The tumbler-dryer shall be constructed so that access for operation, maintenance, and repair shall be analyzed in terms of the applicable requirements as specified in ASTM F1 166. Maintenance and operational activities shall permit performance by the 5th percentile female to the 95th percentile male as specified in ASTM F1166. Access to each control, lubrication fitting, or any other component shall include space for the tools and test equipment required, as well as any protective clothing likely to be worn by the maintainer. Particular attention shall be given to the ease of access for maintenance and removal of the lint box and screens, belt or chain drive and the heating unit. Skill level and training requirements for operation, maintenance and repair shall be kept at the lowest level consistent with other requirements herein. The analysis of operation, maintenance and repair functions shall include and reflect considerations of safety.

3.4 Metal joints. Rivets, bolts, machine screws, nuts, and washers shall be made of steel, except when brass or corrosion-resistant material is fastened, in which case they shall be of brass or corrosion-resistant material. Where dissimilar materials are fastened, rivets, bolts, machine screws, nuts, and washers shall be galvanically compatible with materials being Joined. Electodes or welding rods used for welding shall deposit material similar to the base metal. Nuts subject to vibration shall be adequately locked. Drill point screws shall be prohibited. Self-locking hexagonal head locknuts that will not deteriorate in temperatures up to 250 'F shall be acceptable, except for connecting the heating chamber assembly to the cylinder. Heating chamber connections shall employ steel bolts, nuts, and lockwashers. Lockwashers shall be reusable.

3.5 Details of units or parts Units or parts shall be as specified m 3.5.1 through 3.5.16.4.

3.5.1 Frame. The frame shall be constructed to provide rigid and durable support for housing, cylinder. heating unit, fan motor, and driving mechanism.

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3.5.2 Housing. The entire unit shall be enclosed in a housing finished for corrosion resistance. The housing enclosing the machine shall be of air-tight construction except for inlet, outlet, access plate, and door openings. When in place, doors and access plates shall be tight-fitting. Doors or access plates shall be provided and sized to ensure adequate accessibility for maintenance and repair of machine (see 3.3.8).

305.3 Sweep sheets. The sweep sheets shall be cylindrical in shape and constructed of corrosion-resistant material sheets. Access plates shall be provided in the front and rear for removing lint from between the cylinder basket, sweep sheets, and the housing (see 3.3.8).

3.5.4 Door. A loading door concentric with the cylinder opening shall be provided at the front of the housing. The door shall be insulated and configured to act as a baffle for retaining the load within the cylinder. The door shall be constructed of the same material as the housing and shall be durably hinged. The door shall be provided with a handle and a means of latching when closed and at a minimum of 135 degrees from the closed position. The use of magnetic securing devices shall be prohibited. An electric interlock shall be provided to prevent rotation of the cylinder when the door is opened. The latching mechanism shall be constructed to allow a person trapped within the cylinder basket to push the door open.

3.5.5 Cylinder basket. The cylinder basket shall be formed from perforated galvanized sheet steel (see 3.2.5). The back sheet of the cylinder basket shall be solid, and the front sheet of the cylinder basket shall have a circular opening to match the protruding section of the door. The cylinder shall be driven by a steel cylinder drive shaft welded to a steel transmitter, which shall be securely bolted to the rear cylinder head. The cylinder shall be fitted with lifting ribs equally spaced around the cylinder

3.5.6 Transmitter. The transmitter shall be made of steel with spiders extending to the full cylinder radius and shall be securely bolted to the back sheet of the cylinder and welded to the cylinder drive shaft.

3.5.7 Cylinder drive shaft. Cylinder drive shaft shall be made of a hard, strong steel to ensure safe and durable operation that will withstand ship-induced vibrations. Drive shaft shall be machined for proper fit with bearing sleeves at back of machine.

3.5.8 Cylinder bearings. The cylinder shall be supported at the rear by two heavy-duty roller bearings. The bearings shall be readily renewable and of sufficient capacity and durability to provide rotation of the cylinder under full load conditions without vibrating for not less than 5,000 hours

3.5.9 Lint box and screens A box shall be fitted with a corrosion-resistant screen through which all of the air shall be exhausted The screen shall be installed between the cylinder and the fan. The screen shall he removable, cleanable from the front of the machine. and accessible, as specified m 3,3.8. Screen shall be similar or equal to No. 14 wire mesh size.

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3.5.10 Fan. A fan shall be provided with the capacity to meet the drying requirements specified in 3.6.

3.5.11 Air ducts. Air ducts shall be of sufficient volume to meet the requirements specified in 3.6.1.

3.5.12 Exhaust. An exhaust duct opening shall be provided for exhausting air after passage through the tumbler and filter. The opening shall be provided with fittings for connecting to a ship's ventilation system exhaust air duct.

3.5.13 Drive adjustments. Means shall be provided for adjusting belt or chain drives. Means for adjustment shall be readily accessible as specified in 3.3.8.

3.5.14 Heating units. Heating units shall be as specified in 3.5.14.1 and 3.5.14.2.

3.5.14.1 Heating unit, type I, steam heated. The heating unit shall consist of a continuous tube with closely spaced fins for efficient heat transfer. Coils shall be suitable for a steam working pressure of 100 pounds per square inch (lb/in^2) gauge and provided with l-inch female taper pipe thread size inlet and outlet connections The heating unit shall be constructed in accordance with the ASME Boiler and Pressure Vessel Code, section VIII. The heating unit shall have the capacity to meet the requirements specified in 3.6.1. The heater shall be located in a position to evenly heat and distribute air entering the cylinder. No leaks or distortion shall be permitted throughout the system (see 4.4.3.9 and 3.2.3.1)

3.5.14.2 Heating unit type II, electrically heated. Heating elements shall be configured into an assembly within a chamber. Element shaping shall be done by the heating element manufacturer. Each element shall be fastened securely at one end and rigidly supported or fastened at the other. Each heating element's terminal ends shall be hermetically sealed to prevent moisture infiltrating through the dielectric. Terminal ends to the beating elements shall project a minimum of 1 inch out from the heating chamber to ensure safe electrical connections. Heating element assembly shall be easily removable from the front of the tumbler-dryer and as specified in 3.3.8. The heating element assembly shall be integrated with the temperature controls to form a system that will meet the requirements specified in 3.3.7.5 and the drying efficiency requirements specified in 3.6.1. The heating element assembly power requirements shall not exceed 36 kilowatts (kW) when operated in the high temperature mode and shall balance a three phase load. The dielectric strength of the element shall withstand twice the line voltage plus 1,000 volts without the element shorting (see 4.4.3 .10.1). The cold and hot insulation resistance shall be a mimimum of 25 and 10 megohms, respectively (see 4.4.3 .10.2 and 4.4.3.1 0.3).

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3.5.15 Insulation of housing The outer housing, door and other components contacted by the operator shall be insulated from internal heat sources so that not more than 15 percent of the surface of the outer housing exceeds a temperature of 35 'F above the ambient air temperature (see 4.4.3.12). Insulation shall be encased to hold the insulation in plain and to prevent settling crumbling, or flaking within the housing. The insulation material characteristics shall conform to 3.24.

3.5.16 Electrical requirements. The tumbler-dryer shall operate on a 440 volts alternating current (Vat), 3-phase 60 hertz (Hz) power supply. Unless otherwise specified herein, all electrical equipment shall conform to the recommended practices for electrical installation on shipboard as required by IEEE Standard 45. Electrical equipment shall operate on a type I electrical power system as specified in DOD-STD-1399, section 300. The control panel shall be dripproof m accordance with MIL-STD-108. Pushbuttons and switches shall be arranged for safe and efficient operation in accordance with 3.3.7 and ASTM F1 166. Motors of 1-horsepower size and larger shall not be installed in single-phase circuits. Electrical equipment shall be constructed to operate satisfactorily and without hazard to personnel or equipment with accidental grounds on conductors.

3.5.16.1 Electric motors. Electric motors shall be of dripproof construction and operate the equipment under all operating conditions without exceeding the permissible temperature rise, current rating. Motor bearings shall have adequate facilities for lubrication, Motors shall have proper starting operating characteristics for the driven equipment. Motors shall have the following characteristics in accordance with MIL-M-17060:

Service	C
Bearings	Ball
Insulation	Class B or F
Voltage	440 Vat, 3-phase. 60 Hz
Cooling	Natural
Type	Squirrel cage induction
Horsepower	As required
Ambient temperature	104 'F
Enclosure	Dripproof
Duty	Fan motor: Continuous
	Drive motor: Continuous reversing
	Maximum 4 reversals per minute
	Minimum 1 reversal per minute

3.5.16.1.1 Motor wire. Motors shall be wound with wire sizes conforming to the American Wire Gauge (AWG). Half size wire shall not be used.

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3.5.16.2 Controllers. Motor controllers shall be of the magnetic type with overload protection relays. Motor controllers shall be provided with overload protection for motors 1/8-horsepower and larger. Control circuits shall operate at 115 volts. Reduced voltage control circuits shall be protected by cartridge fuses sized not larger than 200 percent of the transformer kilovoltampere (kVA) rating or 100 percent of the control wire amperage rating, whichever is smaller. Control buttons and switches shall be dripproof in accordance with MIL-STD-108. Indicator lights for voltages in excess of 28 volts shall be of the transformer type with low voltage lamps. Electrical components mounted outside the control panel shall be provided with dripproof enclosures in accordance with MIL-STD-108. Solenoid valves shall be provided with high temperature epoxy encapsulated coils. The control equipment shall have the following characteristics:

Ambient temperature	104 °F
Enclosure	Dripproof
Operation	Manual or magnetic
Type	Ac: across-line
Function	To suit equipment
Duty	To suit equipment

3.5.16.3 Wiring methods and practices. Wiring shall be in accordance with MIL-E-917 and this specification, Wiring between different devices mounted on the outside of the tumbler-dryer shall be in rigid, oil-resistant metal conduit or liquid-tight, flexible metal conduit not smaller than 1/2-inch electrical trade size. Fittings and junction box connections for the conduit entrance shall be either threaded bushings or watertight, neoprene-bushed, strain-relief type. Drainage holes shall be provided at the low point of all conduit runs or junctions. Wiring shall be with moisture-resistant thermoplastic-insulated copper conductors rated for an operating temperature of 221 “F. Power conductors shall be not smaller than No. 12 AWG. Control conductor shall be not smaller than No. 14 AWG. Conductors shall be stranded. When wires are run through holes in partitions, the holes shall be equipped with grommets for mechanical protection of the insulation. Care shall be exercised in running wires to ensure that they are not carried over or bent around any sharp comers or edges. Terminal blocks shall be plainly marked to correspond to markings on the electrical diagram. Solder lugs, soldered joints, and wire-nut connections shall not be permitted. Electrical connections shall be made using solderless ring or pressure fit connectors to parts and terminal boards. Not more than three connections shall be made at each terminal. Conduit, fittings, and wire shall conform to and be installed in accordance with the National Electric Code. The tumbler-dyer shall be used on an ungrounded electrical system, and no grounds shall be permitted on the machine wiring. Connections shall be made between motors, controllers, timers, push buttons, and limit switches installed on the unit. Wires run m conduit or in harness assemblies shall be marked every 12 inches to correspond to the electrical diagram Leads shall be extended from motors and control devices to a connection box and properly identified [or connections to wiring beyond the unit. When mounted on the unit, tumbler-dryers with manual across-the-line starters shall be provided with wiring between the motor and the starter. Electric cable length shall be kept to a minimum, yet have sufficient slack to accommodate machine movements and repair Wiring within the control enclosure shall be neatly arranged into harnesses and secured

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with wire tie wraps. Cable clamps shall be used as necessary to locate harnesses and prevent movement that could cause rubbing or chaffing damage. The electrical contactor and wrong boxes shall be provided with hardware to tightly secure it closed (key latches and magnetic latches shall not be allowed), but will provide for a future access for preventative maintenance and repair. The access door shall be sized so that maintenance tasks shall easily and safely be completed by the fifth to ninety-fifth percentile person, and consideration for the use of tools or test equipment shall be taken (see 3.3.8).

3.5.16.4 Circuit breakers. A properly sized fuse disconnect box shall be installed on the 440 Vac electrical supply line. Properly sized fuses shall also be installed on the secondary side of the stepdown transformer to protect the tumbler-dryer control circuitry.

3.6 Performance characteristics. Performance characteristics shall be as specified in 3.6.1 through 3.6.11.

3.6.1 Drying efficiency. Type I and type XI tumbler-dryers with steam pressure at 100 lb/in^2 and 440 Vat, respectively shall remove moisture from the maximum load capacity soaked with water to a wet weight of 150 percent of the maximum bone-dry weight at a temperature of 70 ± 2 "F. The tumbler-dryer shall remove 97 percent of the water from the load with a drying rate of 1.2 pound of water per minute on type I and 0.8 pound of water per minute on type 11 tumbler-dryers set at a drying temperature of 185 'F (see 4.4.3.2).

3.6.2 Reversing. Reversals per minute shall be set to provide the maximum water removal per minute within the following range. The reversing cylinder shall reverse within a range of four reversals per minutes maximum to a minimum of one reversal per minute (see 4.4.3.3).

3.63 Vibration. Tumbler-dryers shall meet the applicable requirements specified in MIL-STD-161-1 for type I environmental vibration. Acceptability shall be contingent upon the ability of the tumbler-dryer to perform its function during and after testing. Steam heated tumbler-dryers shall only be functionally tested after completion of each test for each plane of motion. Minor damage or distortion shall be permitted during testing providing it does not impair the tumbler-dryer from performing its principle functions. A major failure is one that would cause poor operation or malfunction of the tumbler-dryer or presents a safety hazard. Nonrepetitive failures of parts such as wiring or solenoid valves that may easily be replaced or repaired shall be considered minor failures. Major and minor failure classification shall be left to the discretion of the test engineer (see 4.4.3.8).

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3.6.4 Shipboard power. Tumbler-dxyers shall meet the following shipboard requirements (see 4.4.3.6):

Requirement	Performance criteria
Steady-state voltage	Tumbler-dryer shall operate satisfactorily.
Power interruption	Tumbler-dryer shall return to normal operation without deterioration in performance when power is restored.
Leakage current	Tumbler-dryer under normal operation shall have a leakage current not greater than 5 milliamperes (mA).
Power factor	The tumbler-dryer overall power factor shall exist in the range of 0.8 lagging to 0.95 leading under average voltage tolerance conditions,
Load imbalance	The tumbler-dryer kVA unbalance during normal operations shall not exceed 5 percent.
Insulation resistance	The tumbler-dryer electrical components shall have a minimum cold insulation resistance of 10 megohms.

3.6.5 Shipboard inclined operation. Tumbler-dryers shall function in accordance with the requirements of this specification, without reduction in performance, when inclined at 12 degrees (see 4.4.3.1).

3.6.6 Saturated steam. Pipes fittings coils, and chambers using steam shall be suitable for 100 lb/in^2 saturated steam and shall show no leakage (see 4.4.3.9).

3.6.7 Electromagnetic compatability (EMC) requirements. The tumbler-dryers shall conform to the applicable electromagnetic interference requirements of class A4, as specified in MIL-STD-461 for CE01, CE03, and RE02 (see 4.4.3.7).

3.6.8 Finished painted surface requirements. The finished painted surface requirements shall be as specified in 3.6.8.1 through 3.6.8.3

3.6.8.1 Salt spray requirements. There shall be no evidence of blilstering, creepage, or loss of adhesion of primed panels. Creepage shall not extend more than 1/8 inch on either side of scribed marks. The total width of gap in the prime coat shall not exceed 1/4 inch (see 4.4.3.13.1).

3.6.8.2 Hardness requirements. The surface finish shall remain undamaged (see 4.4.3 13.2).

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3.6.8.3 Adhesion requirements. Normally, no paint squares shall be lifted but a classification of one panel out of two shall be permissible (see 4.4.3.13.3).

3.6.9 Electric heating element strength (type II tumbler-dryers). The electric heating elements on type II tumbler-dryers shall withstand a dielectric voltage of twice the rated line voltage plus 1,000 volts without shorting (see 4.4.3 .10.1). The cold resistance shall not exceed a resistance of 25 megohms (see 4.4.3.10.2). The hot resistance shall not exceed a resistance of 10 megohms (see 4.4.3. 10.3). The elements shall be within plus or minus 5 percent of the rated wattt density for unit length (see 4.4.3.10.4).

3.6.10 Electric motors. Electric motors shall demonstrate their ability to perform and shall not exceed their maximum rated current draw (see 4.4.3.4).

3.6.11 Operation Each tumbler-dryer type and class shall perform all its operations satisfactorily (see 4.4.3.5). Operation of the tumbler-dryer shall not burden an operator within the fifth to ninety-fifth athropedic dimensions by requiring excessive reaching for controls and visually identifying settings or gauge readings.

3.7 Finish and color. Exposed surfaces of the tumbler-dryer shall be painted with a quality heat resistant paint of a color normally furnished on the manufacturer's commercial machine and shall meet the requirements of 3.6.8.

3,8 Identification plate. Identification plated shall be made for the tumbler-dryer and its major components; for example, electric motors and so forth. Identification plates shall be made of corrosion-resistant steel, aluminum alloy, or brass. Characters shall be not less than 3/16 inch m height. Characters shall be stamped on the plate. Stamping shall be not less than 0.003 inch deep. The thickness of the plate shall be 0.03 inch minimum. Plate information shall remain permanent and legible for the life of the equipment.

3.8.1 Mounting. The identification plate shall be securely affixed to each tumbler-dryer in a conspicuous place.

3.8.2 Identification plate data. Data marked on the identification plate shall include the following:

a. Manufacturer's type or model number, serial number, contract or order number, and date of manufacture

b. Salient design features including size, voltage, current, type of heating unit (electric or steam), and capacity in pounds (dry weight per load)

C. Name of tumbler-dryer and stock number (the stock number will be provided to the contractor)

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d. Component identification number (CID). The CID number entry shall consist of the letters CID, followed by a nine digit number that will be furnished upon receipt of provisioning technical documentation. No equipment shall be shipped without the CID number for the tumbler-dryer and its associated components shall also be recorded on the packing list.

3.9 Safety. Exposed parts, such as belts or chains, that may cause injury to personnel shall be covered with adequate guards or shields. Such guards shall be readily removable where maintenance is required. An airflow sensor and temperature sensors shall prevent the cycling of cylinder heating should the blower motor, drum motor fails, or the airflow is restricted. The exhaust and cylinder temperature shall not exceed 250 'F before automatic shutdown (see 4.4.3.11) and the tumbler-dryer shall not scorch its load. For type II tumbler-dryers, the electric heating elements shall not exceed their maximum operating temperature.

3.10 Workmanship. The construction fabrication, and finish shall be suitable for the purpose intended and shall be of the highest quality commensurate with the intended use. Tolerances shall be in accordance with good commercial standards. In general, workmanship shall be equivalent to the best in the industry and shall conform in detail to the following

a. Welds shall be cleaned and ground to present a uniform and smooth finish.

b. Methods of fastening components in place shall provide the strength and rigidity necessary to resist local stresses, internally or externally induced vibration, and impact forces (see 4.4.3.8). Fasteners shall provide the degree of demountability required for maintenance and repair that is not provided by removable or hinged accesses.

c. Rotating parts shall be aligned and balanced in order to prevent vibration and eccentric motion. With regard to cylinder rotation, this requirement shall apply to both empty and loaded conditions.

d. Surfaces and the cylinder interior shall be free of burrs and sharp edges.

e. Chain and belt drive adjustment mechanisms shall be readily accessible and effective as specified in 3.3.8.

f. Access plates and doors shall be provided as necessary for proper maintenance, cleaning, and lubrication and shall be a suitable size and properly located.

g Safety guards shall be adequate and suitably located.

h. Lubrication fittings shall be fitted where necessary and adequate for the purpose intended.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government- The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program (see 6.3). The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of the manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)
b. Quality conformance inspection (see 4.5).

4.2.1 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified herein.

4.3 Intermediate visual and dimensional examination. The intermediate product or assemblies that cannot be checked or measured after final assembly shall be examined at various stages of manufacture for conformance to this specification. Examination shall be made at an appropriate time for defects in construction, workmanship, dimensions, and material. Any deviation from the specified requirements shall be classified as a defect,

4.4 First article inspection. First article inspection shall consist of the examinations shown in table 11 and tests specified m 4.4.2 and 4.4.3.

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TABLE II. Classification of defects.

Examine	Defect (requirement paragraph)	Classification	
		Major	Minor
Materials	Construction materials not as specified (see 3.21 through 3.2.8)	X	
Configuration	Configuration not as specified (see 33.1 through 33.8)	X	
Temperature control	Not as specified (see 33.7.2)	X	
Time drying	Not provided, not as specified (see 33.7.3)	X	
Cooldown phase	Not provided, not as specified (see 33.7.4)	X	
Capacity	Machine not constructed for 50- or 100-pound dry weight capacity (see 33.2)	X	
Mounting	Mounting not as specified (see 333)	X	
Volumetric dimensions	Machine dimensions not as specified (see 33.4)	X	
Weight limit	Machine weight not as specified (see 3.3.5)	X	
Sectionalization	Sectionalizat ton not as specified if required (see 33.6)	X	
Controls	General control not as specified (see 33.7 through 33.75)	X	
Constuction mechanical	Metal joints not as specified (see 34)	X	
	Fasteners not as specified (see 34)	X	
	Frame not as specified (see 3.5.1)	X	
	Housing not as specified (see 3.5.2)	X	
	Sweep sheets not as specified (see 35.3	X	
	Door not as specified (see 3.5.4)	X	
	Cylinder basket not as specified (see 3.	I X	
	Transmitter not as specified (see 3.5.6)	X	
	Cylinder drive shaft not as specified (see 35.7)	X	
	Cylinder hearings not as specified (see 3.5.8)	X	
	Lint box and screens not as specified (see 3.5 9)	X	
	Exhaust not as specified (see 3.5.12)	X	

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TABLE II. Classification of defects - Continued

Examine	Defect (requirement paragraph)	Classification	
		Major	Minor
	Drive adjustments not as specified (see 3.5.13)	X	
	Heating unit not as specified for type I tumbler-dryers (see 3.5.14.1)	X	
	Insulation not as specified (see 3.5.15)	X	
	Finish and color not as specified (see 3.7)		X
Construction, electrical	Heating unit not as specified for type II tumbler-dryers (see 3.5.14.2)	X	
	Electrical requirements are not as specified (see 3.5.16)	X	
	Motors not as specified (see 3.5.16 1)	X	
	Controllers not as specified (see 3 5.16 2)	X	
	Wiring methods and practices not as specified (see 3.5.16.3)	X	
Safety features	Belts, shafts, pulleys not properly enclosed (see 3 9)	X	
	Door latch not provided, or not as specified (see 3.5.4)	X	
	Temperature limit controls not as specified (see 3.3.7.5)	X	
Identification	Identification plate not of specified material (see 3.8)		X
	Markings illegible (see 3.8)	X	
Workmanship	Surfaces of cylinder not free of burrs and sharp edges (see 3.10(d))	X	
	Access plates and doors not as specified (see 3.10(f))	X	
	Welds not ground (see 3.10(a))		X
	Lubrication fittings not as specified (see 3 10(h))	X	

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4.4.1 First article sample. One tumbler-dryer of each design and size shall be submitted for first article inspection.

4.4.2 Contractor tests. Before submitting the first article sample for inspection, the contractor shall subject the machine to the test specified in 4.4.3.5.

4.4.3 First article inspection tests. First article inspection tests shall consist of the trots as specified in 4.43.1 through 4.4-3.13.3.

4.43.1 Inclination. The tumbler-dyer shall be inclined at an angle of 12 degrees to each side of two vertical planes at right angles to each other and shall operate satisfactorily in each of the four positions for 30 minutes (see 3.6.5).

4.4.3.2 Drying efficiency. Test cloths shall be prepared from a white, cotton fabric of uniform texture and permanent finish, having a warp of 48, a welt of 48, and weighing not less than 0.33 pound per square yard. Permanent finish, sanforized, white cotton cloth, known as Indianhead cloth may be used. The test cloths shall be cut 24 by 36 inches and be double-hemmed to a finished size of 22 by 34 inches. An average of 10 runs shall be used to evaluate the drying efficiency of the machine (see 3.6.1).

4.4.3.3 Overall construction and performance. The tumbler-dryer shall be tested to determine conformance to the following requirements:

Cooldown	- Automatic to 135 'F (see 3.3.7.4)
Reversing	– 4 reversals per minute (maximum) 1 reversal per minute (minimum) (see 3.6.2)
Door interlock	- (see 3.5.4)

4.43.4 Power input After accelerating to full speed under maximum loaded conditions, the voltage received by the motor shall be measured to determine if it is within 5 percent of the rated value (see 3.5.16.1).

4.43.5 Durable press controls. Testing of durable press controls (see 3.3.7) shall be as follows:

a. With the tumbler-dryer loaded to capacity and operating at a maximum temperature setting, the thermostatically-umtrolled solenoid valve on type I and electric heater thermostat on type 11 tumbler-dryers shall limit the temperature to the prescribed settings.

b. The tumbler-dryer shall be operated for 10 complete cycles at 185 and 155 "F. The thermostatically-controlled solenoid valve on type I tumbler-dryers and thermostat

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temperature sensors on type 11 tumbler-dryers shall consistently maintain a cylinder temperature within plus or minus 10 "F of the selected temperature. The operation of the drying timer (see 3.3.7.3), timer integrated cooldown (see 3.3.7.4), indicating lights (see 33.7.3 and 3.3.7.4), temperature indicator (see 3.3.7.2), and audible signal (see 3.3.7.4) shall be verified.

4.4.3.6 Electrical tests. Tumbler-dryers shall be subjected to the electrical tests specified in 4.4.3.6.1 through 4.4.3.6.6. Tumbler-dryers shall meet the requirements specified in 3.6.4.

4.43.6.1 Steady-state voltage and frequency. The tumbler-dryer shall be operated for at least 15 minutes at the following five conditions:

440 Vac supplied equipment

440 Vat, 60 Hz
418 Vat, 58 Hz
462 Vat, 58 Hz
462 Vat, 62 Hz
418 Vat, 62 Hz

4.4.3.6.2 Power interruption. With the tumbler-dryer operating within the steady-state voltage and frequency tolerances specified in 4.4.3.6.1, the external power supply shall be suddenly interrupted. After an interval of 0.5 to 20 seconds, the power supply, within the steady-state tolerances, shall be reapplied and the tumbler-dryer restarted. The test shall be repeated four times.

4.4.3.6.3 Leakage current. With the frame of the tumbler-dryer ungrounded and the equipment operating within the steady-state voltage and frequency tolerances specified m 4.4.3.6.1, the leakage current for each phase between the frame shall be successively measured through a 1,500 ohm resistive load.

4.4.3.6.4 Power factor. With the tumbler-dryer operating within the steady-state voltage and frequency tolerances specified in 4.4.3.6.1, the voltage, current, and wattage shall be measured and the power factor computed as follows:

Three-phase

$$\text{Power factor} = \frac{\text{Wattage}}{\text{Voltage average} \quad \text{*Current average} \quad \text{*(3 ** 1/2)}}$$

4.43.6.5 Load imbalance. With the tumbler-dryer operating at 440 Vac-60 Hz, the current on each phase shall be measured and the load imbalance computed as follows:

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$$\text{Load imbalance} = \frac{I_{maximum} - I_{minimum}}{I_{average}}$$

4.43.6.6 Dielectric strength test. A dielectric test voltage of 1,500 Vac-60 Hz having a capacity of at least 1 kW shall-be applied successively between each phase line and the equipment frame. 'The voltage shall be raised gradually and shall be maintained for 1 minute plus or minus 5 seconds. The insulation resistance between all mutually insulated points and between all insulated points and the frame shall be measured with a 500 volt direct current (Vdc) megohmeter.

4.43.7 Electromagnetic interference tests. MIL-STD-461. class 4A, tests CE03, CE03, and RE02 shall be performed on each machine. Graphical limiting emissions for surface ships shall be as specified in MIL-STD-461. Test setups shall be as specified in MIL-STD-462 (see 3.6.7).

4.43.8 Vibration test. Tumbler-dryer shall meet the vibration test requirements specified m MIL-STD-167-1 for type I equipment m a frequency range of 4 Hz to 25 Hz (see 3.6.3).

4.4.3.9 Hydrostatic test. The steam coils and piping systems on type I tumbler-dryers shall be hydrostatically tested to a pressure of 150 lb/in^2 gauge (which is 150 percent of the maximum system operating pressure) for a minimum of 30 minutes. No leaks or distortion in the system shall be permitted (see 3.6.6).

4.43.10 Electric heating element tests. Electric heating element tests shall be as specified in 4.4.3.10.1 through 4.4.3.10.4.

4.43.10.1 Dielectric strength test. A dielectric test voltage of twice the rated line voltage plus 1,000 volts shall be applied between the terminal and the sheath (see 3.6.9). The frequency of the test voltage shall be 60 Hz, root mean square, alternating current (at), and shall approximate q true sine wave. The test shall be conducted with the heating elements at the ambient room temperature.

4.4.3 .10.2 Insulation resistance, cold test for electric heating elements. The insulation resistance shall be measured with a 500 Vds insulation resistance tester with the elements at room temperature (see 3.6.9).

4.43.103 Insulation resistance, hot test for electric heating elements. The insulation resistance shall be measured with a 500 Vdc insulation resistance tester 5 seconds after the operation of the element at the maximum operating temperature of the sheath material end seal material, or resistance wire material, whichever is largest. The ambient fluid shall not be circulated past the element after the power is removed (see 3.6.9).

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4.43.10.4 Wattage. The wattage draw for the heating element shall be measured (see 3.6.9).

4.43.11 Limiting devices. One No. 24 AWG bead type thermocouple shall be mounted in the exhaust outlet, drying cylinder of the tumbler-dryer and another on the top center row of the heating coils or heating elements. The heating elements or coils shall not exceed their maximum operating temperature. Exhaust temperature or cylinder temperature shall not exceed 250 "F during test runs. After one full operation at the high temperature setting, the tumbler-dryer shall be placed into operation at the maximum temperature setting with a full load, as specified in 4.4.3.2. The tumbler-dryer shall be placed into the following operational conditions until automatic shutdown:

a. Failure of blower motor
b. Failure of drum rotation
c. Failure of drum rotation and blower motor
d. Failure of drum rotation, blower motor, and airflow switch.

The time of shutdown and the temperature of the thermocouples shall be recorded. Upon conclusion of each failure mode, a visual inspection of the test load shall be conducted (see 3.9).

4.4.3.12 Surface and handle temperature test Each exterior surface of the tumbler-dryer shall be divided into 6 by 6 inch equal areas on each panel. Surface temperatures shall be measured at the center of each area, and the areas explored for the maximum temperature. The temperatures of all exterior surfaces contacted by the operator shall also be measured; for example, handles, latches, controls, and so forth. Temperature measurements shall be taken by an indicator potentiometer and an iron-constantan surface temperature thermocouple, having its junction brazed on the face of a copper plate 0.22 inch in diameter and 0.024 inch in thickness. 'he iron- constantan lead wire from the junction shall be No. 24 AWG. The tumbler-dryer shall be loaded with three test loads as specified in 4.4.3.2. Each load shall be dried at the maximum temperature setting and the maximum time. Just before the thermostat functions for the third load, the temperature readings shall be taken (see 3.5.15).

4.4.3.13 Painted surface testing. Three test panels of the cabinet wall thickness and composition shall be paint finished to withstand testing for adhesion, hardness, and salt spray resistance. Two panels shall be prime coated only and two panels shall be prime coated and painted in the same way that the cabinet of the tumbler-dryer would be perfected at the factory. The test panels shall be a width of 4-1/2 ± 1/2 inches and a length of 7-1/2 ± 1/16 inches.

4.4.3.13.1 Salt spray test- Salt spray test shall be conducted in accordance with ASTM B 117 and the following. Two primed panels shall be scribed on one side of each panel with two diagonal scribe lines extending through the prime coat down to the base metal. For scribing, a sharp instrument and a straight edge shall be used. The scribing tool shall be held perpendicular to the surface of the test panel to make two intersecting, diagonal scribed marks. Each scribed line shall begin at a point approximately 1/2 inch from a corner and continue to a point approximately 1/2

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inch from the diagonally opposite corner. The scribe shall penetrate the prime coating down to the metal, leaving a uniformly bright line Each metal panel shall be exposed to a salt fog Concentration of 5 ± 1 percent with the panel perpendicular to the spray. Before the salt spray fog, the panels shall be conditioned in the chamber for 2 hours. The panels shall be exposed for 240 hours. The salt spray fog shall be emitted from an orifice diameter size of 0.02 plus 0.03 inch that atomizes approximately 2.8 listers of salt solution per 10 cubic feet of chamber volume in a 24-hour period. The temperature of the chamber shall be maintained at 95 ± 2 'F, and the salt spray shall be maintained within plus or minus 10 "F of the chamber temperature. Each test panel shall be removed from the salt spray cabinet and rinsed off using a gentle stream of water at a temperature of 100 ± 10 "F. The test panels shall be dried off, and the scribed sides of each panel shall be scraped with a dull knife blade or a piece of sheet metal having a similar dull edge. The scraper shall be held with its face perpendicular to the specimen surface and parallel to the scribed lines, moving it back and forth across the scribed lines. The scraping shall be completed within 15 minutes of the drying of the surface. Creepage or loss of adhesion shall be measured at the scribed marks after scraping. Masking tape 1-inch wide shall be pressed onto all the scribed marks and pulled away clearly (see 3.6.8.1).

4.43.13.2 Hardness test. Two primed and painted panels shall be subjected to a hardness test in accordance with ASTM D 3363 using a standard 2H drafting pencil, sharpened to expose the full diameter of the pencil lead. The lead shall be sanded to a blunt-end with sharp edges and shall be shaped by using sanding strokes perpendicular to the centerline of the lead. Holding the end of the lead firmly against the enamel test panel with the pencil centerline at 45 degrees on the painted surface, fifteen gouging or scraping strokes shall be applied to the enamel. The strokes shall be a minimum of 1/4-inch long and applied forward to gouge the lead into the enamelled surface (see 3.6.8.2).

4.4.3.13.3 Adhesion test. Adhesion between the paint and primer and between the primer and base metal of the panel shall be evaluated by a crosshatch test m accordance with ASTM D 3359 and the following. The test panels shall be scribed with a sharp knife or similar instrument to give a crosshatch pattern with an overall size of 1 square inch. Seventeen parallel cuts shall be made at about 1/19-inch intervals extending through the coating to the base metal. Seventeen similar cuts shall be made at 90 degrees crossing the first seventeen, producing a crosshatched area of 256 squares. Masking tape l-inch wide shall be pressed firmly onto the entire hatched area and then immediately jerked cleanly away from the panel. The hatched area shall then be examined (see 3.6.8.3).

4.5 Quality conformance inspection. Quality conformance inspection shall be as specified m 4.5.1 through 4.5.3.3 (see 6.3).

4.5.1 inspection lot. A lot shall consist of all tumbler-dryers of one design offered for delivery at one time.

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4,5.2 Sampling for visual and mechanical inspections. Examination of the end item shall be for defects specified in table II.

4.5.3 Sampling for operational tests. The lot shall be selected for the tests specified m 4.5.3.1 through 4.5.3.3. If any machine fails any test, the entire lot shall be rejected

4.53.1 Operational testing. Each sample tumbler-dryer shall be connected to power and operationally tested without garments or laundry. The following items and operations shall be checked:

a. Reversing of cylinder (see 3.3.1 and 3.6.2)
b Manual selection of temperatures (see 3.3.7.2)
c. Selection of drying times (see 3.3.7.3)
d. Operation of cooldown phase (see 3.3.7.4)
e. Door interlock (see 3.5.4)
f. No undue vibration, noise, or overheating
g. Operator to machine operation (see 3.6.11)
h. Activation of the airflow switch with a reduction in airflow capacity through tumbler-dryer (see 3.9).

4.5.3.2 Power input. The power input to each motor shall be measured when the motor is operating at its maximum mechanical load (see 3.6.10).

4.533 Insulation testing An insulation resistance test shall be conducted on each test tumbler-dryer, motor, controller, and connecting cable. The tests shall use a 500 Vdc insulation resistance tester. Insulation resistance for each component shall be not less than 10 megohms (see 3.6.4).

4.6 Inspection of packaging. Sample packages and packs, and the inspection of the preservation, packing and marking for shipment, stowage, and storage shall be m accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.2).

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5.1 General

5.1.1 Navy fire-retardant requirements.

a. Treated lumber and plywood. Unless otherwise specified (see 6.2), all lumber and plywood including laminated veneer material used in shipping containers and pallet construction shall be fire-retardant treated material conforming to MIL-L-19140 as follows:

Levels A and B	Type 11- weather resistant Category 1 - general usc
Level C	Type I – non-weather resistant Category 1 - general use.

b. Fiberboard. Fiberboard used m the construction of class-domestic, non-weather resistant fiberboard, cleated fiberboard boxes including interior packaging forms shall meet the flame spread index and the specific optic density requirements of PPP-F-320 and amendments thereto.

5.2 Preservation, packing, and marking. Tumbler-dryers shall be preserved level A or commercial packed, level A, B or commercial, as specified (see 6.2) and marked in accordance with MIL-L-3153. Tumbler-dryers shall be shipped fully assembled, unless disassembly is necessary to protect certain components during transit, disassembly results in more than a 25 percent savings m transportation costs to the government, or the sectionalized sections exceed Volumetric constraints (see 3.3.6) plus shipping container and packaging.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but IS not mandatory.)

6.1 Intended use. Laundry tumbler-dryers acquired under this specification are intended for shipboard use. Therefore, commercial machines offered to the Government may require alteration to meet specification requirements; for example, mounting provisions, electrical power, and materials.

6.2 Acquisition requirements. Acquisition documents must specify the following

a. Title, number, and date of this specification

b. Classification: type, class, and style (see 1.2)

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c Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2)

d. When first article is required (see 3.1)

e. When fire-retardant materials are required (see 5.1.1)

f. Level of preservation, packing and marking required (see 5.2).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
4.1.1, 4.5	DI-T-2072	Reports, test	—

The above DID's were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6,4 Technical manuals. The requirement for technical manuals should be considered when this specification is applied in a contract. If technical manuals are required, military specifications and standards that have been cleared and listed in DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL) must be listed on a separate Contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract.

6.5 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first lot production items, a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified m 4.3. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for sample.. for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish

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evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so m the solicitation.

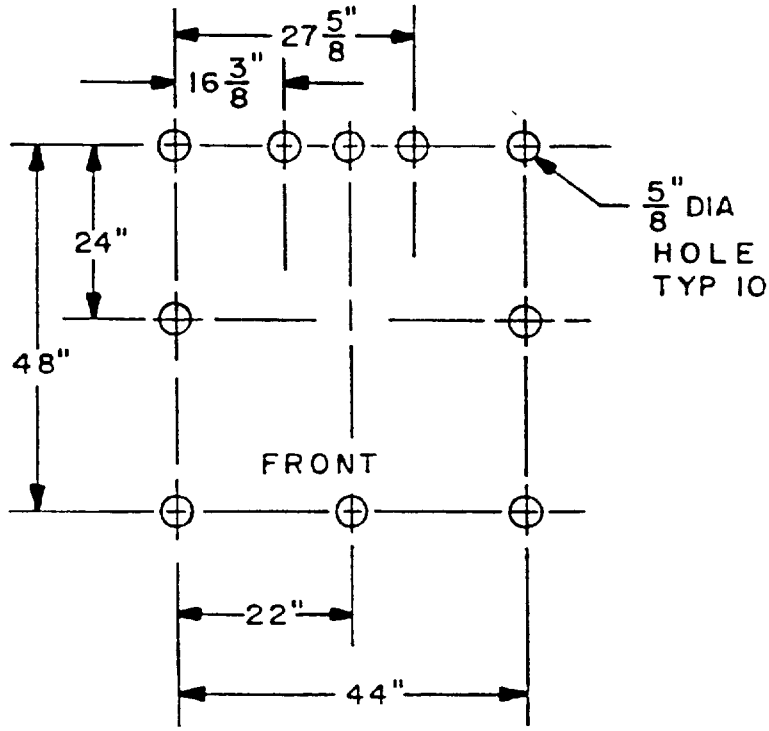
6.6 Subject term (key word) listing.

Cooldown
Dripproof
Housing insulation
Reversing
Shutdown
Temperature control

6.7 Changes from previous issue. Marginal notations arc not used in this revision to identify changes with respect to the previous issues due to the extensiveness of the changes.

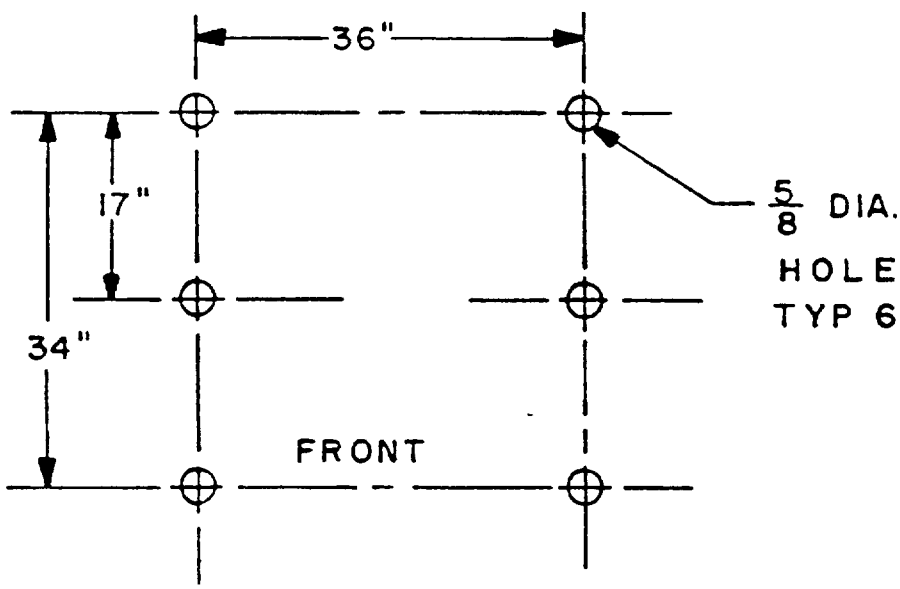
Preparing activity:
Navy - SH
(Project 3510-N305)

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SH 13231745

FIGURE 1. *Tumbler-dryer deck mounting.*



SH 13231746

FIGURE 2. *Tumbler-dryer deck mounting.*

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-T-23480A(SH)	90/09/27

DOCUMENT TITLE
TUMBLER-DRYER, LAUNDRY, STEAM AND ELECTRIC MODIFIED COMMERCIAL (SHIPBOARD)

NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

REASON FOR RECOMMENDATION

SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
ADDRESS (Include Zip Code)	c. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

PREPARING ACTIVITY

NAME	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON
Technical Point of Contact (TOPC): Mr. Dick Atkins (SEA 55W51) PLEASE ADDRESS ALL CORRESPONDENCE AS FOLLOWS:	TPOC: 703-602-1591
ADDRESS (Include Zip Code) Commander, Naval Sea Systems Command Department of the Navy (55Z3) Washington, DC 20362-5101	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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